Band-1 Computational Experiment: Erdos Discrepancy - Pi_1 Finite C=1 (tau(1)=12)

Human + ZoraASI full-dash (executor)

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Band-1 Computational Experiment: Erdos Discrepancy — Π_1 Finite $C = 1$

Honesty stamp. This note does **not** claim the full Erdos discrepancy problem (Tao 2015) is re-proved on this box. It does **not** re-derive the SAT certificate $\tau(2) = 1160$ of Konev–Lisitsa. It closes **only** the classical finite case $\tau(1) = 12$ by exhaustive search with an independently checkable certificate. **GATE-04 collection remains LOCKED.** No Clay / ToE stamp.

Abstract

For a ± 1 sequence $x = (x_1, \dots, x_L)$, the *homogeneous discrepancy* is

$$\mathrm{disc}(x) = \max_{d=1,\dots,L} \Big| \sum_{i=1}^{\lfloor L/d \rfloor} x_{id} \Big|.$$

Define $\tau(C)$ as the least n such that every ± 1 sequence of length n has disc $> C$. We close by exhaustive enumeration

$$\tau(1) = 12.$$

There are exactly **4** sequences of length 11 with disc ≤ 1 (**2** up to global sign flip), matching OEIS A237695 examples `+--+-++--+-` and `+--+-++--++` (and their inverses). There are **0** such sequences of length 12 (all $2^{12} = 4096$ checked; DFS-with-prefix-prune agrees with brute force for every $L \in \{1, \dots, 12\}$). Wall ≈ 0.04 s.

Problem statement (Erdos-adjacent, finite)

Erdos conjectured that disc is unbounded on every infinite ± 1 sequence (the Erdos discrepancy problem / EDP). The full conjecture was proved by Tao (2015). The finite thresholds $\tau(C)$ remain of independent interest (OEIS A237695): $\tau(1) = 12$ (classical), $\tau(2) = 1160$ (Konev–Lisitsa SAT), $\tau(3)$ open in general (multiplicative case settled at 127645).

Tonight's Π_1 target. The statement

$$\forall\, x \in \{\pm 1\}^{12} :\ \mathrm{disc}(x) \geq 2$$

is a finite universal claim, decidable by search, and is the $C = 1$ case of EDP — **not** twin primes, **not** an open asymptotic.

Literature ledger (survey 2026-09-11)

Item	Status	Source / note
$\tau(1) = 12$	**CLOSED** (classical; box re-certified tonight)	elementary case analysis / OEIS A237695
Maximal disc-≤ 1 length	**11**	OEIS: `+--+-++--+-`, `+--+-++--++`, inverses
$\tau(2) = 1160$	literature	Konev–Lisitsa arXiv:1402.2184
Multiplicative $\tau_\times(3)$	127645 (lit.)	Konev–Lisitsa extended
Full EDP (all C)	**SOLVED** (Tao 2015)	Discrete Analysis 2016 — **not** re-proved here
Formal Conjectures angle	decidable finite fragment	suitable Lean/SAT target

Method (box)

1. **Discrepancy oracle.** For each d, accumulate $x_d + x_{2d} + \cdots$ and track absolute partial sums; take max over d.
2. **Brute force.** For each $L \leq 12$, enumerate all 2^L sequences; count those with disc ≤ 1.
3. **DFS with prefix prune.** Grow a partial sequence; prune when the current prefix already has disc > 1. Must agree with brute on counts and survivor lists.
4. **Certificate.** Emit the full list of length-11 survivors, the hash of that list, and the 4096-check that none exist at length 12. OEIS witnesses verified; each extends by ± 1 to disc ≥ 2.

Results (numbers!)

Unit tests

`PASS` (6/6): all-+ length 3 has disc 3; OEIS witnesses disc = 1; sign-flip invariant; length-1 disc = 1; alternating length 12 has disc ≥ 2.

Census: $\#\{x \in \{\pm 1\}^L : \mathrm{disc}(x) \leq 1\}$

L	Count	DFS nodes	Brute match
1	2	3	yes
2	2	5	yes
3	4	9	yes
4	2	11	yes
5	4	15	yes
6	2	17	yes
7	4	21	yes
8	2	23	yes
9	4	27	yes
10	2	29	yes
11	**4**	33	yes
12	**0**	33	yes

Exact claim closed

$\tau(1) = 12$.

- Length-11 survivors (pm notation):
  `+--+-++--++`, `+--+-++--+-`, `-++-+--++-+`, `-++-+--++--`.
  Sign-orbit representatives: the first two (OEIS pair).
- Length 12: **none** (checked $= 4096 = 2^{12}$).
- Dual methods (DFS prune + brute) **agree** for all $L \leq 12$.
- Each OEIS maximal extends by either sign to disc ≥ 2.

What is not claimed

- Full EDP / Tao unboundedness theorem.
- $\tau(2) = 1160$ re-derivation (gigabyte UNSAT certificate; literature only).
- Any new asymptotic discrepancy bound.
- Clay / ToE / GATE-04 unlock.

Reproducibility

```
cd experiments/band1/erdos_pi1
python3 edp_c1_exact.py
# writes results.json, certificate_tau1_eq_12.json, SUMMARY.txt, run.log
```

Independent check of the certificate: recompute `disc` on each of the four length-11 strings (expect 1); for all 4096 length-12 bitmasks, assert `disc >= 2`.

Conclusion

Band-1 / Π_1 Erdos-adjacent target **closed**: $\tau(1) = 12$ by exhaustive search with a clear, tiny certificate. This is a finite decidable fragment of the classical Erdos discrepancy problem — honest scope, not a famous open.

References (selected)

1. P. Erdos, problems on discrepancy of ± 1 sequences (1930s–1950s).
2. OEIS A237695 — Maximum length of a ± 1 sequence of discrepancy n.
3. B. Konev & A. Lisitsa, *A SAT attack on the Erdos discrepancy conjecture*, arXiv:1402.2184 (2014).
4. T. Tao, *The Erdos discrepancy problem*, Discrete Analysis (2016).